

[ACJC/2025/Q1]

1 Name your Jupyter Notebook as:

TASK1_<your name>_<centre number>_<index number>.ipynb

A magic square of order n is an $n \times n$ array of integers that has the following properties:

- Each integer from 1 to n^2 inclusive appears exactly once in the array;
- Each row in the array has the same sum;
- Each column in the array has the same sum;
- The two longest diagonals of the array have the same sum as the rows and columns.

For example, a magic square of order 4 is shown below. Each row, each column, and each of the two longest diagonals has a sum of 34.

16	2	3	13
5	11	10	8
9	7	6	12
4	14	15	1

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 1.1
        Program code
        Output:
```

Task 1.1

Write a function `check_magic(arr)` that takes in an $n \times n$ array `arr` of integers and returns `True` if it forms a magic square, and `False` if it does not. Data validation does not need to be performed. [11]

Task 1.2

Write a function `check_file(filename)` that checks if an array, stored in a file, forms a magic square. It can be assumed that the file contains an $n \times n$ array of integers. [5]

Test your function with the files `Task1-1.csv` and `Task1-2.csv`, showing your output. [2]

Task 1.3

When n is odd, a magic square can be constructed in the following way.

- Start from the centre entry of the top row and insert 1 into the array there.
- Repeat the following steps until all the integers from 1 to n^2 are in the array.
 - Suppose you have just inserted the integer k into the array. The next number to be inserted is $k + 1$.
 - Move one step to the right and one step up. If this brings you outside the array, wrap around to the bottom row or the left column as necessary.
 - If the entry is already occupied by a number, go to the entry directly below the previous number instead, wrapping around to the top row if necessary.
 - Insert $k + 1$ into this entry.

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For example, the first few steps to construct a magic square of order 5 are shown below.

		1		

Step 1: Insert 1 in the centre of the top row

		1		
			2	

Step 2: Move one step right and one step up. Since 1 is in the top row, you need to wrap around to the bottom row. Insert 2.

		1		
				3
			2	

Step 3: Move one step right and one step up. Insert 3

		1		
4				
				3
			2	

Step 4: Move one step right and one step up. Since 3 is in the rightmost column, you need to wrap around to the leftmost column. Insert 4.

		1		
	5			
4				
				3
			2	

Step 5: Move one step right and one step up. Insert 5.

		1		
	5			
4	6			
				3
			2	

Step 6: Moving one step right and one step up brings you to an occupied square. Insert 6 directly below 5.

Write a function `make_magic(n)` that takes in an odd integer `n` and constructs and returns a magic square of order `n`. [6]

Create a magic square of order 9 and save it to `Task1-3.csv`. [5]

Save your Jupyter Notebook for Task 1.